

MVPConnect — External Artist Knowledge Graph

Future Feature Notes / Product + Architecture Checkpoint

Purpose: preserve the current product idea for artist-reference resolution, Spotify-assisted search, asynchronous AI enrichment, and relationship-derived matching signals so the feature can be resumed later without reconstructing the design decisions.

1. Feature Summary

When an MVPConnect user adds an artist reference—Artist “Sounds Like,” Venue “Artists You’ve Booked,” or Promoter “Artists You’ve Worked With”—MVPConnect should resolve that name to a reusable external-artist record. The platform checks its own database first. If the artist is unknown, Spotify can be used as an identity/search source. The selected artist is then created in MVPConnect and marked for asynchronous enrichment.

A scheduled enrichment process (initial idea: approximately every 10 minutes) finds external artists that need enrichment, researches/classifies them, and maps the results into MVPConnect’s canonical genre/vibe taxonomy. Those inferred attributes become matching signals for every MVPConnect user connected to that external artist.

2. Product Relationships

Persona	User-facing concept	Future graph relationship
Artist	Sounds Like	(:Musician)-[:SOUNDS_LIKE]->(:ExternalArtist)
Venue	Artists You’ve Booked	(:Venue)-[:HAS_BOOKED]->(:ExternalArtist)
Promoter	Artists You’ve Worked With	(:Promoter)-[:HAS_WORKED_WITH]->(:ExternalArtist)

These three relationships are deliberately different semantically while sharing the same resolved artist identity. This creates direct cross-persona evidence for matching.

3. Resolution Flow

1. User begins typing an artist name.
2. Search MVPConnect’s ExternalArtist index/database first.
3. If found, use the existing ExternalArtist record.
4. If not found, search Spotify for artist candidates.
5. Show enough identity information for the user to choose the correct artist when names are ambiguous.
6. Create an MVPConnect ExternalArtist using the resolved external identity and mark enrichmentStatus = PENDING.
7. Create the appropriate persona relationship (SOUNDS_LIKE / HAS_BOOKED / HAS_WORKED_WITH).

8. Do not make onboarding wait for AI enrichment; resolution + relationship persistence is sufficient to continue.

4. Spotify's Role

Spotify is a resolver/enrichment source, not MVPConnect's source of truth. MVPConnect should retain its own stable ExternalArtist ID and store Spotify identifiers as external identifiers. If Spotify access or API behavior changes, the MVPConnect graph must remain valid.

Useful Spotify-derived identity fields may include:

- spotifyId
- spotifyUrl / URI
- artist name
- available artist imagery or other permitted catalog metadata

Important constraint noted during planning: Spotify's current Web API documentation says Spotify Platform/Content may not be used to train or otherwise be ingested into an AI/ML model. Therefore, future AI enrichment must respect Spotify's then-current developer terms and should not assume Spotify content can be fed to the enrichment model.

5. ExternalArtist Concept

Initial conceptual fields:

- id (MVPConnect stable ID)
- name
- normalizedName
- spotifyId (optional)
- spotifyUrl (optional)
- spotifyImageUrl or permitted identity-image reference (optional; policy-dependent)
- enrichmentStatus
- enrichmentVersion
- enrichmentAttempts
- processingStartedAt / processingOwner (if lease/claim is implemented)
- lastEnrichedAt
- nextEnrichmentAt
- inferredGenreCodes[]
- inferredVibeCodes[]
- createdAt / updatedAt

6. Enrichment Lifecycle

Proposed status lifecycle:

- PENDING → PROCESSING → COMPLETE

- FAILED → retry later according to nextEnrichmentAt

The first implementation can use a Spring scheduled job roughly every 10 minutes. It should claim a bounded batch of eligible records before processing so multiple application instances cannot enrich the same artist concurrently. A stale PROCESSING lease should eventually be recoverable.

If enrichment volume later grows, the same domain model can move to a queue/worker architecture (for example SQS or Kafka) without changing the user-facing relationship model.

7. AI Enrichment Contract

The enrichment agent should classify into MVPConnect’s canonical taxonomy rather than inventing arbitrary final profile tags. Candidate evidence can come from permitted public/trusted sources. The system should retain provenance and confidence where practical.

Conceptual enrichment evidence:

- taxonomy code (e.g., PUNK, RAW)
- confidence
- source/evidence category
- classifier/enrichment version
- timestamp

The enrichmentVersion field is important: when the classifier or taxonomy improves, older ExternalArtist records can be marked for re-enrichment deterministically.

8. Explicit vs. Inferred Signals

Do NOT silently write inferred ExternalArtist tags into a user’s self-selected genres/vibes. User-declared taxonomy and reference-derived inference are different kinds of evidence.

Signal	Example	Treatment
Explicit	Artist selected INDIE + PUNK	Canonical user profile; strongest evidence
Reference relationship	Artist SOUNDS_LIKE IDLES	Strong relationship evidence
Reference-derived inferred	IDLES classified RAW / GRITTY / HARDCORE	Weaker matching evidence; do not rewrite user profile

Matching weights are intentionally undecided. The important principle is that explicit user choices, direct shared artist references, and inferred reference-derived tags remain distinguishable so they can be weighted differently.

9. Matching Opportunities

The graph creates both direct and semantic evidence. Example: an Artist SOUNDS_LIKE an external artist, a Venue HAS_BOOKED that same artist, and a Promoter HAS_WORKED_WITH that artist. This direct shared reference is meaningful even before enrichment. Once the ExternalArtist has inferred genres/vibes, MVPConnect gains additional semantic overlap signals.

- shared ExternalArtist reference
- explicit genre overlap
- explicit vibe overlap
- event-type overlap
- reference-derived genre overlap
- reference-derived vibe overlap
- later: location, draw/capacity, travel, availability and other operational fit

10. Demand-Driven Knowledge Graph

Do not pre-populate millions of artists. Build the ExternalArtist graph on demand from artists that MVPConnect users actually reference. The first reference resolves/creates/enriches the artist; subsequent references reuse the same record and enrichment result. Re-enrichment occurs only when required (for example, a newer enrichmentVersion).

11. Step 2 / Onboarding Relationship

Current Step 2 onboarding can remain simple and free-form for V1. The existing EntityReferenceDto is a precursor to this feature. Today an external reference may be stored with entityType=ARTIST, entityId=null, displayName, and external=true. The future resolver can replace free-form references with stable ExternalArtist IDs without changing the user-facing questions.

The enrichment feature is explicitly deferred. Step 2 V1 should not wait for Spotify integration or the AI agent.

12. Duplicate / Identity Rules to Preserve

- If a resolved entity ID exists, identity should be based on entity type + stable ID.
- For unresolved/free-form references, duplicate comparison should use normalized display name (trim, case-insensitive, collapse repeated whitespace).
- Preserve the user-friendly display spelling separately from normalization used for comparison/search.
- Spotify IDs should be treated as external identifiers, not MVPConnect primary IDs.

13. Future Search UX

Artist-reference entry can evolve from free-form input to a resolver-backed search. Search MVPConnect first; if no good result exists, query Spotify. Ambiguous names should show selectable candidates. Free-form fallback can remain available if external resolution fails, depending on future product decisions.

The same resolver can power Artist “Sounds Like,” Venue “Artists You’ve Booked,” and Promoter “Artists You’ve Worked With.”

14. Separate Spotify Use Case: Artist's Own Profile

Connecting an Artist's own Spotify account/catalog for profile music playback/showcase is a separate relationship from using Spotify to resolve a "Sounds Like" reference. The two features may share provider infrastructure but should remain distinct in the domain model.

15. Deferred Decisions

- Exact ExternalArtist Neo4j schema and whether inferred tags remain properties or become taxonomy relationships.
- Exact enrichment evidence/provenance storage model.
- Which non-Spotify public/trusted sources the AI agent may use.
- Exact AI provider/model and enrichment prompt/agent architecture.
- Exact scheduler batch size, retry/backoff policy and stale-lease timeout.
- Matching weights for explicit vs. direct-reference vs. inferred signals.
- Whether/when enrichment-derived signals are materialized per user versus calculated from graph traversal.
- Spotify authentication/quota-mode strategy and then-current developer-policy review.
- ExternalArtist merge/deduplication and alias handling.
- Admin/review tools for incorrect enrichment.
- Re-enrichment/migration behavior when MVPConnect taxonomy changes.

16. Guiding Principles

- MVPConnect owns the durable artist identity and matching taxonomy.
- External providers help resolve/enrich identities; they do not become the durable domain source of truth.
- Enrich once, reuse many times.
- Do enrichment asynchronously; never block onboarding on the AI agent.
- User-declared tags remain user-declared; inferred signals remain visibly/architecturally distinct.
- Store provenance/versioning so inferred intelligence can be explained, audited and recomputed.
- Build the knowledge graph from actual user demand rather than crawling the entire music universe up front.

17. Resume Point

When this feature is resumed, begin by reviewing current Spotify API capabilities/terms, then design the ExternalArtist node + relationship schema and artist-resolution API. After that, implement MVPConnect-first search, Spotify fallback resolution, asynchronous enrichment, and finally integrate inferred/reference signals into the matching engine with separately tunable weights.